

Ex. No.: 17

Date:

AIM

To implement Functions using programs in MySQL for the Hostel Management System database.

HIGH LEVEL PROGRAMMING EXTENSIONS - FUNCTIONS

FUNCTIONS

A function is a stored program that computes and returns a value. In MySQL, a user-defined function is created using CREATE FUNCTION and can be called using SELECT.

Creating a Function

The CREATE FUNCTION statement is used to create a stored function. The function accepts input parameters and returns a value of a specified data type.

MySQL Stored Function Syntax

```
CREATE FUNCTION function_name(parameter)
RETURNS datatype
DETERMINISTIC
BEGIN
    -- statements
    RETURN value;
END;
```

Problem 1

Create a function that returns the room level based on the monthly rent from the Hostel_Rooms table. If the rent is 5000 or more, the room level is PLATINUM. If the rent is 4000 or more and less than 5000, the level is GOLD. Otherwise, the level is SILVER.

Program-1

```
USE HostelDB;

DROP FUNCTION IF EXISTS GetRoomLevel;

DELIMITER //

CREATE FUNCTION GetRoomLevel(p_roomid INT)
```

```

RETURNS VARCHAR(10)
DETERMINISTIC
BEGIN
    DECLARE v_rent DECIMAL(8,2);
    DECLARE v_level VARCHAR(10);
    SELECT RentPerMonth
    INTO v_rent
    FROM Hostel_Rooms
    WHERE RoomID = p_roomid;
    IF v_rent >= 5000 THEN
        SET v_level = 'PLATINUM';
    ELSEIF v_rent >= 4000 THEN
        SET v_level = 'GOLD';
    ELSE
        SET v_level = 'SILVER';
    END IF;
    RETURN v_level;
END //

```

```

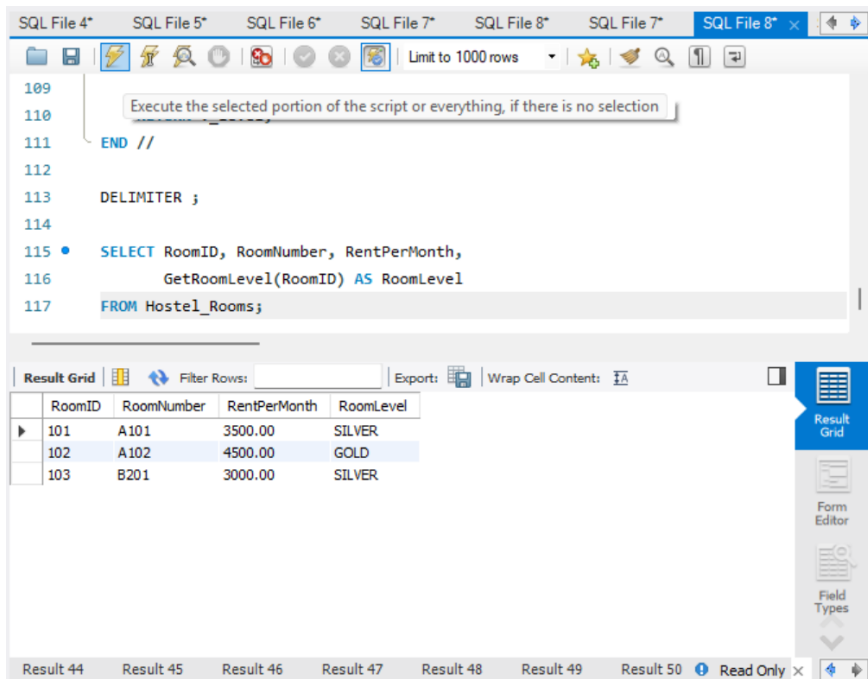
DELIMITER ;

```

```

SELECT RoomID, RoomNumber, RentPerMonth,
       GetRoomLevel(RoomID) AS RoomLevel
FROM Hostel_Rooms;

```



The screenshot shows a SQL IDE interface with multiple tabs at the top labeled 'SQL File 4*' through 'SQL File 8*'. The active tab is 'SQL File 8*'. The script editor contains the following SQL code:

```

109
110
111 END //
112
113 DELIMITER ;
114
115 • SELECT RoomID, RoomNumber, RentPerMonth,
116       GetRoomLevel(RoomID) AS RoomLevel
117 FROM Hostel_Rooms;

```

Below the script editor is a 'Result Grid' showing the output of the query. The grid has four columns: RoomID, RoomNumber, RentPerMonth, and RoomLevel. The data is as follows:

RoomID	RoomNumber	RentPerMonth	RoomLevel
101	A101	3500.00	SILVER
102	A102	4500.00	GOLD
103	B201	3000.00	SILVER

The IDE also features a toolbar with various icons for file operations, a 'Limit to 1000 rows' dropdown, and a 'Filter Rows' field. The bottom status bar shows 'Result 44' through 'Result 50' and a 'Read Only' indicator.

Problem 2

Write a recursive MySQL function to compute the factorial of a number.

Program-2

```
USE HostelDB;
```

```
DROP FUNCTION IF EXISTS Factorial;
```

```
DELIMITER //
```

```
CREATE FUNCTION Factorial(n INT)
```

```
RETURNS BIGINT
```

```
DETERMINISTIC
```

```
BEGIN
```

```
    IF n <= 1 THEN
```

```
        RETURN 1;
```

```
    ELSE
```

```
        RETURN n * Factorial(n - 1);
```

```
    END IF;
```

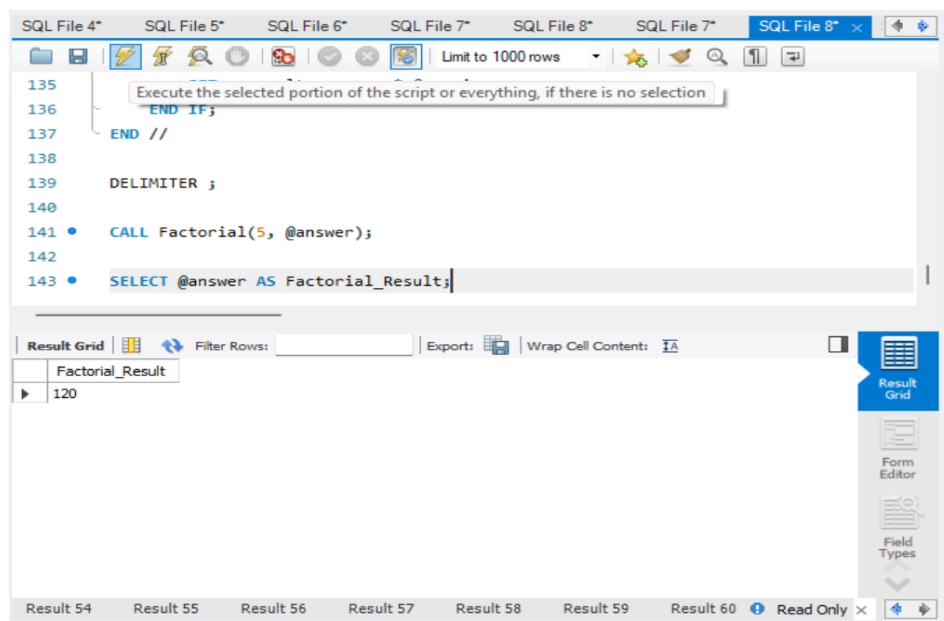
```
END //
```

```
DELIMITER ;
```

```
SELECT Factorial(5) AS Factorial_Result;
```

Expected Output

Program-1 displays the RoomID, RoomNumber, RentPerMonth and corresponding RoomLevel. Program-2 displays 120 as the factorial of 5.



RESULT

Thus, the Functions using programs in MySQL were executed successfully using the HostelDB database.